

Tuesday Reading Assessment: Unit 0, Review of 135A

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1 Memory Bank

- $\vec{a} = a_x\hat{i} + a_y\hat{j}$... Component notation for 2D vector
- $\vec{a} + \vec{b} = (a_x + b_x)\hat{i} + (a_y + b_y)\hat{j}$... Adding vectors
- $\vec{a} - \vec{b} = (a_x - b_x)\hat{i} + (a_y - b_y)\hat{j}$... Subtracting vectors
- $|\vec{a}| = \sqrt{a_x^2 + a_y^2}$.. Magnitude of a 2D vector
- $a_x = |\vec{a}| \cos \theta$... x-component of a 2D vector
- $a_y = |\vec{a}| \sin \theta$... y-component of a 2D vector
- **Coulomb's Law:** Let two charges with magnitudes q_1 and q_2 be separated by a displacement $\vec{r}$. The force between them is

$$\vec{F} = \frac{kq_1q_2}{r^2} \hat{r} \quad (1)$$

The vector $\hat{r}$ is a unit vector in the direction of $\vec{r}$, and r is the magnitude. The constant k is $k = 8.988 \times 10^9 \text{ N m}^2 \text{ C}^{-1}$.

- $\vec{F} = q\vec{E}$... The electric field $\vec{E}$ causes the Coulomb force on a charge q .
- **Coulomb Field:** Let a charge with magnitude q be separated by a displacement $\vec{r}$ from a *test charge*. The E-field generated by q is

$$\vec{E} = \frac{kq}{r^2} \hat{r} \quad (2)$$

The vector $\hat{r}$ is a unit vector in the direction of $\vec{r}$, and r is the magnitude. The constant k is $k = 8.988 \times 10^9 \text{ N m}^2 \text{ C}^{-1}$.

2 Warm-Up Exercises

1. Perform the following unit conversions:

- Convert 320 gm to kg:
- Convert $2.4 \times 10^5 \text{ cm s}^{-1}$ to km hr^{-1} :
- Convert the density of ice, $0.9167 \text{ gm cm}^{-3}$, to kg m^{-3} .

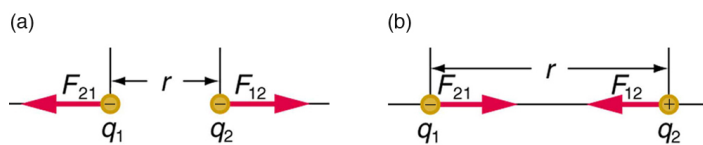


Figure 1: Coulomb forces between charges.

- Let $\vec{x} = 0.5\hat{i} - 0.5\hat{j}$, and $\vec{y} = -0.5\hat{i} + 0.5\hat{j}$. (a) Calculate $\vec{x} + \vec{y}$. (b) Calculate $\vec{x} - \vec{y}$. (c) What is the magnitude of $\vec{a}$? (d) What is the magnitude of $\vec{b}$?
- Let $\vec{F}$ be a 10 N force that makes a 60 degree angle with the x-axis. (a) What is the x-component of this force? (b) What is the y-component? (c) Suppose this force displaces a system 3 m along the x-axis. What is the work done? *Recall that $W = \vec{F} \cdot \Delta\vec{x}$.*
- Suppose a charge $q_1 = -1 \text{ nC}$ is located 1 cm away from a charge $q_2 = +1 \text{ nC}$. What is the magnitude and direction of the force between the charges? To interpret the direction, consider Fig. 1.
- Suppose four equal charges q form a square with side length a . What is the magnitude of the electric field at the center of the square?